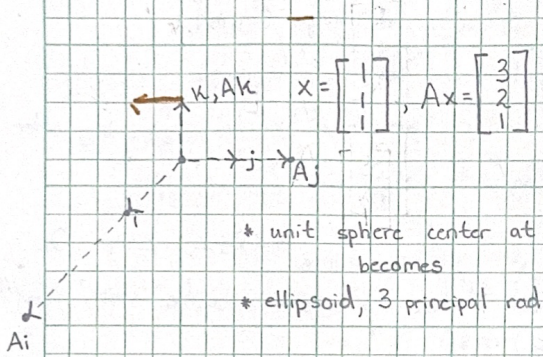
3×3 SVD ex:

$A = U\Sigma V^T$

$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ – transforms vectors in 3D-space



$x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $Ax = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

* unit sphere center at origin becomes
* ellipsoid, 3 principal radii = 3, 2, 1

1) calculate A^TA

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

↓ how transformation changes vector lengths before final orientation

for any vector x:

$\|Ax\|^2 = x^TA^TAx$ → how much length a vector will change under A, depending on direction

$$14 = (1\ 1\ 1)\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$(3\ 2\ 1)\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(9\ 4\ 1)\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 14\ \checkmark$$

2) eigen → $\det(A^TA - \lambda I) = 0$

$$\det\begin{pmatrix} 9-\lambda & 0 & 0 \\ 0 & 4-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{pmatrix} = 0$$

$$9-\lambda\begin{vmatrix} 4-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0$$

$$9-\lambda(\lambda^2 - 5\lambda + 4) = 0$$

$$(9-\lambda)(\lambda-4)(\lambda-1) = 0$$

$\lambda = 1, 4, 9$

3) singular values = $\sigma_i = \sqrt{\lambda_i}$

$\sigma_1 = 3, \sigma_2 = 2, \sigma_3 = 1$

$\Sigma = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ → largest to smallest

5) cols of V unit eigenvectors of A^TA

$\lambda_1 = 9 \rightarrow \begin{pmatrix} 0 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -8 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$0v_1 + 0v_2 + 0v_3 = 0,\ 0 = 0,\ v_1 = 1$

$0v_1 + -5v_2 + 0v_3 = 0,\ v_2 = 0$

$0v_1 + 0v_2 + -8v_3 = 0,\ v_3 = 0$

$\lambda_2 = 4 \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\lambda_3 = 1 \rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

which cols of V tell us which directions in original input space correspond to principal stretch or compress.

6) U → each nonzero singular value, calculate corresponding left singular vector using $u_i = Av_i/\sigma_i$

for $\sigma_1 = 3$, $u_1 = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\sigma_2 = 2$, $u_2 = \frac{1}{2}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$,

$\sigma_3 = 1$, $u_3 = \frac{1}{1}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

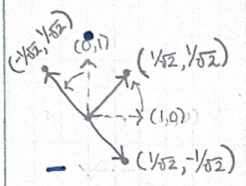
↓ final directions of transformed principal axes

(remain aligned w/ original coordinate axes)

directions on which scaling or compressing happens (rotation 45° from I)

suppose: $V = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$ ↳ always measure from I standard basis

V col, rotated principal directions



$V^T = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$

$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

— Vx

— $V^T x = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.41 \\ 0 \end{pmatrix}$ ↗ from standard basis

↓

(1.41) along $(\underbrace{1/\sqrt{2}}_{x}, 1/\sqrt{2})$ = → original vector points at 45°, exactly along V_1, when we express in rotated coordinate system

$.7071 + (.7071 * .41) =$

$.7071 + .2899 = .997 \approx 1$

1) component along $V_1 = 1.41$

2) component along $V_2 = 0$

(0) along $(-1/\sqrt{2}, 1/\sqrt{2}) = 0$

(1.41, 0) along $\begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$ gives you (1,1) in correspondence standard basis $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$